

Citrus Yield Estimation Method Based on Video Stream and YOLOv8n Object Tracking

Naijin Si^{a,b}, Jianrong Cai^{a,*}, Li Sun^{a,c}, Chen Wang^a, Wenhui Zhu^a, Bingke Pan^a, Zhiguo Shi^a

^aSchool of Food and Biological Engineering, Jiangsu University, Zhenjiang, China

^bSchool of Agricultural Engineering, Jiangsu University, Zhenjiang, China

^cSchool of Mechanical Engineering, Jiangsu University, Zhenjiang, China

* Corresponding author. School of Food and Biological Engineering, Jiangsu University, Xuefu Road, Jingkou District, Zhenjiang City, Jiangsu Province, China, 212013

E-mail addresses: jrcai@ujs.edu.cn (J. Cai), sinaijin_765@163.com (N. Si)

Abstract

Accurate citrus yield estimation is crucial for fruit yield control and quality enhancement. An automated citrus orchard yield estimation model was developed by capturing full-surface video streams of individual fruit trees during their enlargement stage using a rotating camera system. This model employs a YOLOv8n-SC detector integrated with the DeepSORT multi-object tracking algorithm. The impact of various image data on estimation accuracy was investigated under distinct illumination conditions: Sunny conditions, Overcast conditions, and Artificial illumination. Different shooting angles were also examined: Overhead shooting, Horizontal shooting, and Low-angle shooting. The YOLOv8n model was enhanced by incorporating a small object detection head to reduce feature information loss. The CBAM attention mechanism was introduced into the C2f module to strengthen extracted features and positional data. Additionally, the WIoU loss function replaced CIoU to improve model convergence performance. Results indicate YOLOv8n-SC achieved a recall rate of 97.6% and a mean average precision mAP_{50} of 93.7% on the total dataset, outperforming the original YOLOv8n model by 2.0% and 3.5% respectively. Optimal detection performance for YOLOv8n-SC was observed under Artificial illumination, with an mAP_{50} of 98.2%. Under Artificial illumination with Horizontal shooting, tracking accuracy reached 90.8% and tracking precision 94.0%. Average counting accuracy was 89.46% with a mean absolute error of 9.98. These tracking and counting data were more stable compared to Overhead shooting and Low-angle shooting. This estimation system effectively counts individual citrus fruits, providing a valuable technical reference for citrus yield prediction.

Keywords :

Citrus development stage, Rotational imaging, Yield assessment, YOLOv8n-SC, DeepSORT, Agricultural automation

1. Introduction

As the global leader in cultivation area and production, citrus is a pivotal economic crop in China, significantly impacting farmers' livelihoods. Presently, citrus yield estimation is primarily conducted via manual sampling, a process recognized as laborious and time-consuming [1], consequently failing to meet contemporary industry demands. Advancing citrus yield estimation towards automation is therefore considered highly significant for enhancing agricultural production efficiency and promoting smart agriculture [2].

Object detection algorithms are considered fundamental for accurate yield estimation [3], enabling precise fruit identification, localization, and stable tracking within complex environments. Traditional object detection methods, such as image segmentation [4] and color segmentation [5], are often compromised by environmental interferences like occlusion and varied illumination, limiting their generalizability [6]. Conversely, deep learning-based object detection [7], utilizing convolutional neural networks, offers advantages in terms of lightweight architecture, scalability, and high speed, rendering it more suitable for real-time applications. He and Xiong [8] proposed a green fruit detection method named deep bounding box regression forest (DBBRF) to address the challenge of detecting green citrus in natural environments, achieving a mAP_{50} of 87.6% with an average execution time of 0.759 seconds. Riza and Musahada [9,10] demonstrated that the original YOLOv7 model outperforms its tiny and X counterparts in citrus fruit detection, achieving an mAP_{50} of 90.6% along with strong precision and sensitivity metrics. Lu and Chen achieve 93.6% accuracy in detecting green citrus fruit in natural environments using a YOLOv5 backbone network [11], surpassing mainstream lightweight models while reducing parameters and model size. Wu proposes the automatic fruit counting system [12], which utilizes YOLOv4-tiny and threshold-based abatement of abnormal fruit detection to improve fruit counting accuracy from 89.1% to 95.0% in modern apple orchards.

The escalating scale of agricultural production coupled with diminishing rural labor has amplified the urgency for automated orchard yield estimation. However, existing image-based estimation methods have been found to be insufficient for high-precision counting [13]. Consequently, a video stream-based object tracking estimation approach is proposed in this paper.

Traditional object tracking techniques, such as optical flow and frame differencing [14], are often limited by challenges posed by illumination variations and background interference. While LiDAR-based object tracking methods [15] acquire high-precision three-dimensional spatial data capable of capturing target shape and position, they are susceptible to occlusion and noise interference, thus hindering high-precision estimation requirements. Deep learning, characterized by robust generalization, efficient and precise representation, and ease of deployment, facilitates not only automated yield estimation but also intelligent agricultural applications including maturity discrimination [16] and mechanized harvesting [17]. Wang, Walsh, and Koirala proposed a method for on-tree mango fruit detection and counting that combines deep learning, a Kalman filter, and the Hungarian algorithm, achieving a bias-corrected Root Mean Square Error (RMSE) of 18.0 fruits per tree, outperforming traditional dual-view imaging methods [18]. Jin proposes the CO-YOLO model for recognizing the growth patterns of *Camellia oleifera* fruits in 2D images, achieving a precision of 90.6% and improving mAP_{50} by 4.8% compared to YOLO11s while significantly reducing model size and computational complexity [19]. Li proposes a video stream inspection method utilizing the

YOLOv5 network and an improved ByteTrack algorithm to accurately classify and count dragon fruit flowers and fruits, achieving an overall recognition accuracy of 95.0% and a counting speed of 56 frames per second [20].

Existing video data collection predominantly involves inter-row parallel shooting [21,22], a methodology primarily suited for trellis-trained or espaliered fruit trees. Conversely, contemporary fruit tree orchards typically feature uniformly spaced plantings where branches extend radially, resulting in fruit distribution at various heights and orientations within the canopy. Rotating capture around individual trees can provide comprehensive surface image data, thereby mitigating estimation errors arising from fruit occlusion and enhancing the model's detection capability and counting precision. This paper aims to construct an estimation model that integrates deep learning object detection algorithms with multi-object tracking and counting algorithms, thereby facilitating automated citrus counting based on video streams.

2. Materials and Methods

2.1. Data Sources

2.1.1. Data Collection Sites

As citrus growth is dependent on effective accumulated temperature, Jiangsu is not considered an ideal citrus cultivation region; consequently, the fruit enlargement stage is entered later due to its colder climate. Therefore, experiments were commenced in late September at the Hongmeiren greenhouse cultivation base in Wutao Village, Jiangxin Isle, Zhenjiang City, Jiangsu Province (119°58'E, 32°21'N). Video stream data were collected from 20 citrus trees during their fruit enlargement stage. The citrus trees were cultivated using a high-density planting method, characterized by a row spacing of 2.5–3 meters, a plant spacing of 2–2.5 meters, and an approximate plant height of 2 meters.

2.1.2. Data Acquisition Methods

A handheld camera system was constituted by mounting a sports camera and a high-power LED light source onto a telescopic pole (maximum 1.5 meters). Raw video stream data were acquired using a DJI OSMO Action4 sports camera (as shown in Figure 1a).

Full-surface image acquisition of fruit trees was achieved by rotating the handheld camera system 360° around each tree. Experimental conditions encompassed varied illumination conditions (Sunny conditions, Overcast conditions, Artificial illumination) and distinct shooting angles (Overhead shooting, Horizontal shooting, Low-angle shooting) [23]. The camera's distance from the tree trunk's central point (rotation radius) was approximately 1.2 m to 1.5 m, adjusted according to tree width and inter-tree spacing, based on the average canopy height H . Based on height H , Horizontal shooting was defined as horizontal capture [24]. Overhead shooting involved a downward angle of 30°, while Low-angle shooting utilized an upward angle of 30° (as shown in Figure 1b). The acquired data comprehensively covered various illumination conditions, including direct light, back light, side light, and Artificial illumination.

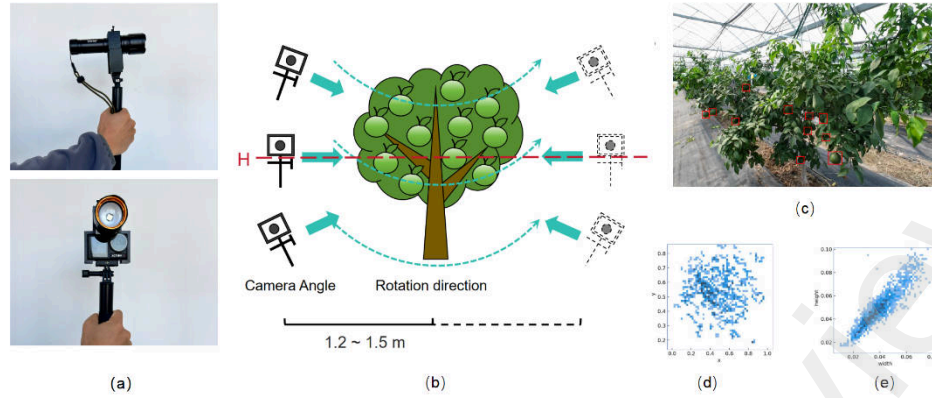


Fig.1-Data acquisition and labeling. (a) Handheld camera system.(b) Data acquisition method.(c) Example of Labeling.(d) Distribution of labels center coordinate.(e)Distribution of labels width and height.

Nine combinations were arranged based on varying illumination and viewing angles, as exemplified in Figure 2. Specifically, 9 videos were captured per tree, totaling 180 videos from 20 fruit trees. All videos were saved in MP4 format with a resolution of 3840*2880 and a frame rate of 30 frames per second. Following video processing to remove redundant captured sections, each video lasted approximately 30 seconds. One image was automatically extracted every 18 frames, resulting in 9,000 citrus images (total dataset). The datasets for Sunny conditions, Overcast conditions, and Artificial illumination each contained 3,000 images.

The AnyLabeling annotation tool was utilized for manual annotation of the dataset in YOLO format. During annotation, minimal bounding boxes were employed to identify citrus targets. Only green citrus fruits with manually discernible or inferable contours on the current tree were annotated. Fallen fruits, rear-row fruits, fruits at image edges, and cracked fruits on the tree were not annotated. Upon completion of annotation, multiple rounds of auditing were conducted to ensure annotation accuracy. The dataset was subsequently randomly divided into training and test sets at a 4:1 ratio.

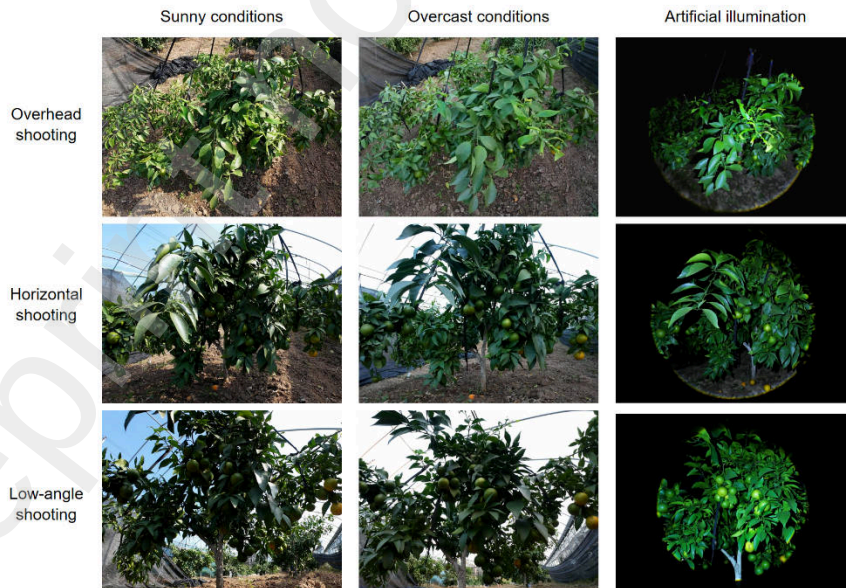


Fig.2-Data examples of nine groups

2.2. Fruit Detection Based on Improved YOLOv8n

YOLO series models are characterized by their compact size [25], strong flexibility, and high accuracy. Object detection tasks can be completed in a single forward pass, which significantly enhances real-time detection processing speed. Nevertheless, false negatives or false positives can still occur when affected by occlusion and illumination variations. To overcome these shortcomings, a small object detection algorithm, YOLOv8n-SC, based on YOLOv8n, was proposed.

2.2.1. Additional Detection Head

In the YOLOv8n model, the head output (Head) is responsible for object detection and classification tasks. By default, three detection heads (Detect) are provided for the detection of large, medium, and small objects [26], respectively. Citrus fruits typically constitute a very small proportion of an image, with their bounding box area often being less than 0.1% of the total image area. This characteristic presents multiple challenges for micro-object detection. Furthermore, issues such as occlusion, blurring, and color distortion contribute to sub-optimal fruit recognition performance by the original detection heads. To address these issues, a new detection head was introduced, specifically optimized for micro-object features with unique anchor boxes and prediction layers [27]. As illustrated in Figure 3, a small object output layer with a 160×160 detection feature map size was incorporated. This detection head is designed to collaborate with the existing three detection heads, thereby forming a more robust detection module. By reducing the receptive field, citrus object detection can be performed in parallel across multiple detection heads [28]. This approach effectively minimizes feature information loss during the downsampling process, consequently reducing instances of missed citrus detections.

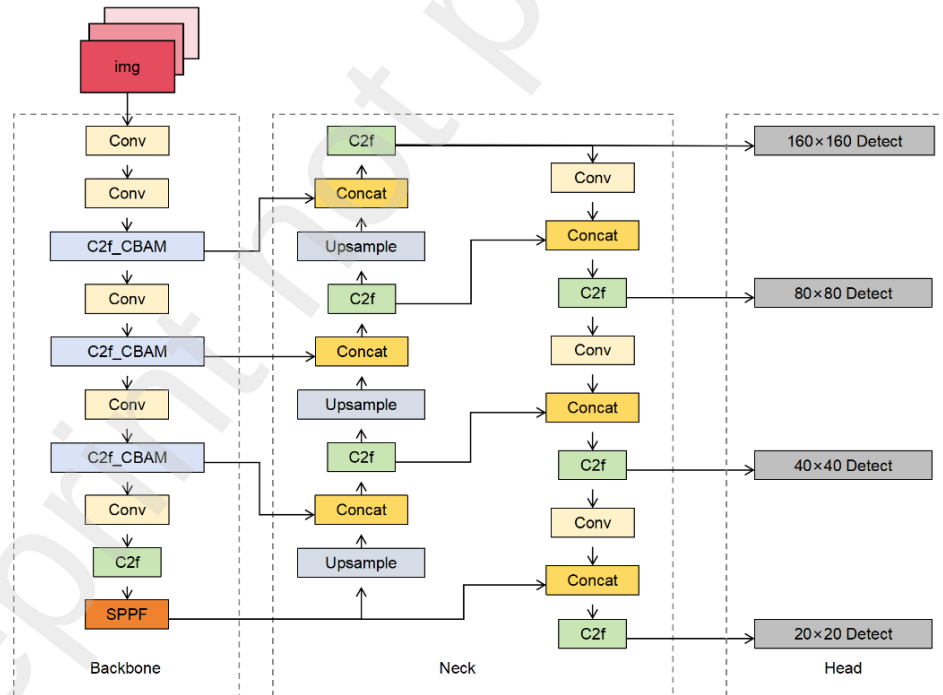


Fig.3 -The structure of YOLOv8n-SC model

2.2.2. CBAM Attention Mechanism

Intersection over Union is illustrated in Figure 5, where α represents the weight parameter and v is utilised to measure the aspect ratio.

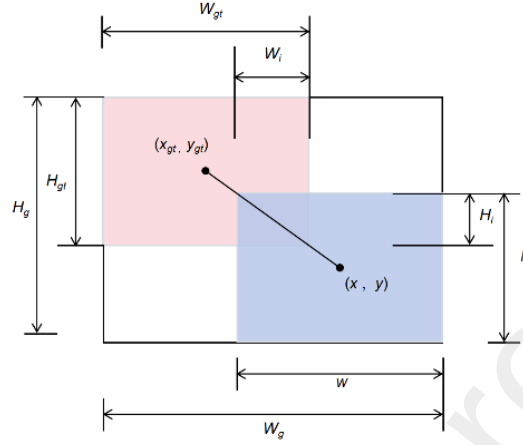


Fig.5-IoU loss function. $w, h, (x, y)$ represent the width, height, and center coordinates of the predicted bounding box; $W_{gt}, H_{gt}, (x_{gt}, y_{gt})$ represent the width, height, and center coordinates of the ground truth bounding box; W_i, H_i represent the width and height of the intersection area; W_g, H_g represent the width and height of the minimum enclosing bounding box.

During the detection process, citrus fruits are characterized by color similarity to the background and severe occlusion, rendering their identification challenging. To enhance precision and accelerate bounding box regression speed, the original loss function is replaced by the Weighted Intersection over Union [22], which incorporates a dynamic non-monotonic focusing mechanism. Its loss function, as presented in Equation (5), defines an outlier parameter β to characterize the abnormality of anchor boxes. Parameters α and δ adjust the relative weight of samples during loss calculation [31]. The non-monotonic focusing coefficient r dynamically adjusts the bounding box gradient gain by prioritizing ordinary quality anchor boxes, thereby reducing the contribution of high-quality samples to the loss value. This mechanism minimizes detrimental gradients generated by low-quality anchor boxes during the later stages of training, ultimately enhancing the model's localization capability.

$$L_{WIoU} = r \cdot R_{WIoU} L_{IoU}, R_{WIoU} \in [1, e), L_{IoU} \in [0, 1] \quad (5)$$

$$r = \frac{\beta}{\delta \alpha^{\beta - \alpha}} \quad (6)$$

$$R_{WIoU} = \exp\left(\frac{(x - x_{gt})^2 + (y - y_{gt})^2}{(c_w^2 + c_h^2)^*}\right) \quad (7)$$

WIoU is designed to eliminate the aspect ratio penalty term present in CIoU, balancing the influence of high-quality and low-quality anchor boxes on model regression. This approach consequently strengthens the model's generalization capability and improves overall performance.

2.3. Fruit Tracking and Counting Based on DeepSORT

2.3.1. State Estimation and Fruit Matching

As the camera rotates circularly, citrus fruits are observed to exhibit varying occlusion states, leading to considerable differences in identified fruit counts even between proximate frames. Consequently, the multi-object tracking algorithm DeepSORT is introduced to facilitate real-time tracking and counting of citrus within complex environments. Its framework structure is presented in Figure 6.

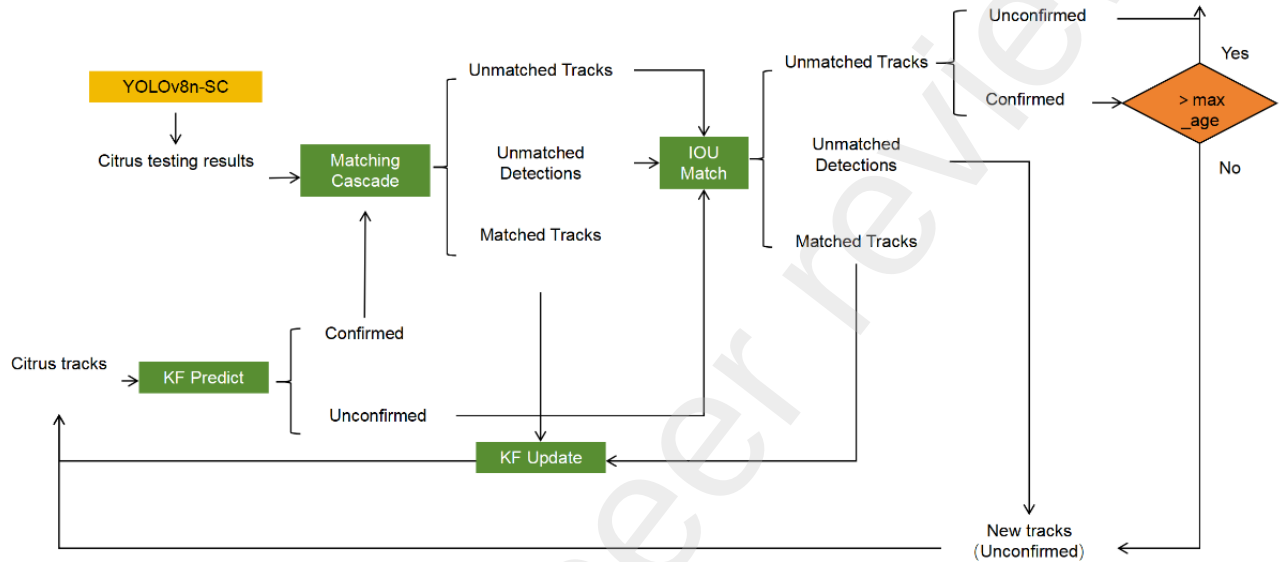


Fig.6-The structure of DeepSORT citrus tracking frame

Given the minimal positional changes of fruit targets between consecutive video frames, their motion can be approximated as uniform. Kalman filtering, an estimation algorithm for linear motion states, enables accurate state estimation through prediction and update cycles [32]. During the prediction stage, the current state is initially estimated using the state and control input from the preceding moment. In the update stage, state estimation is further optimized by integrating actual observations and correcting predictions. This methodology is particularly well-suited for processing noisy data, offering real-time uncertainty management within dynamic systems, thereby enhancing tracking stability and precision.

Association calculation is a critical step for matching detection results with predicted trajectory states. The optimization of DeepSORT is primarily based on the cost matrix derived from the Hungarian algorithm. An additional cascaded matching process is performed prior to IoU matching, which integrates appearance features and Mahalanobis distance, rendering it suitable for object detection in dynamic environments. During target tracking, deviations between predicted and true positions can be effectively quantified by calculating the Mahalanobis distance. When target motion patterns fluctuate frequently or targets become occluded, cosine distance is introduced to capture appearance features, thus maintaining a robust matching capability. These two distinct measurement methodologies serve to complement each other in facilitating accurate target matching.

2.3.2. Tracking and Counting

For each track within the DeepSORT algorithm, an initialized timer (age) of zero is assigned. This counter is incremented by one upon an unsuccessful match and reset to zero following a successful match. The 'age' value signifies the time elapsed since the last successful correspondence for that track. A maximum age (max_age) of 90 frames is pre-defined; should a track fail to be re-matched within this duration, it is subsequently deleted.

The detailed algorithm process, as illustrated in Figure 6, is outlined below:

(1) For each citrus fruit within a video frame, corresponding track information is created. Kalman filter motion variables are initialized, and their associated detection box information is predicted.

(2) Tracks from Step 1, alongside targets detected by YOLOv8n-SC in the current iteration, are submitted for cascade matching. Citrus targets are then classified into three distinct states: unmatched detections, unmatched tracks, and matched tracks.

(3) To address potential omissions from Step 2, these are combined with unconfirmed tracks from Step 1 and re-matched using IoU matching, similarly resulting in three states.

(4) Unconfirmed tracks and those confirmed but with an 'age' exceeding 90 are subsequently deleted from the set of unmatched tracks.

(5) Matched tracks are merged and outputted following a Kalman filter update. Unmatched targets from Step 3 are designated as newly established tracks. Tracks confirmed in Step 4 but with an 'age' less than or equal to 90 are also considered for a new round of tracking [33]. These three categories collectively form the basis for the next tracking iteration.

(6) Steps 1-5 are iteratively executed until the video frame sequence concludes. Numerical IDs, commencing from 1, are assigned to fruits based on their detection order within the video frame, thereby achieving automated citrus counting.

2.4. Citrus Yield Estimation Experiment Design

The experiment was divided into three main sections: object detection, object tracking, and object counting. The specific processing workflow is illustrated in Figure 7.

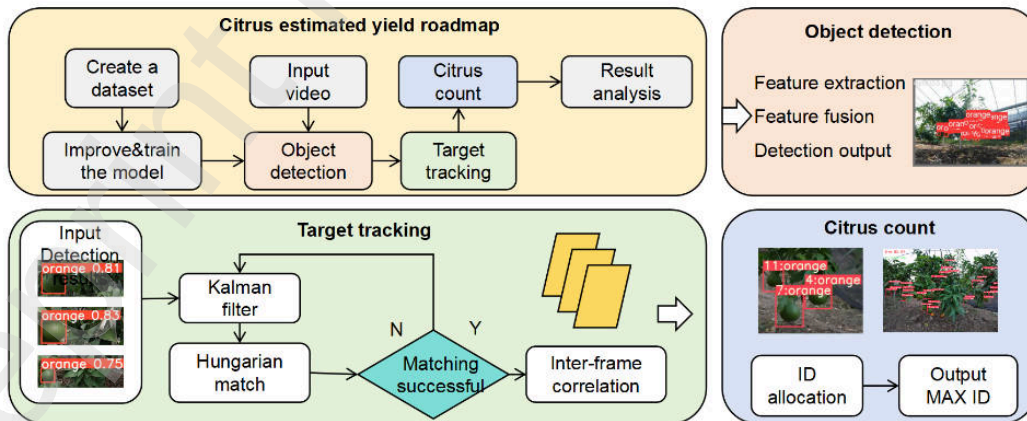


Fig.7-Citrus yield estimation roadmap

The experimental environment comprised an Intel(r) Xeon(r) Platinum P-8124 CPU @ 3.00GHz 3.00 GHz (dual-core) processor, 64 GB of random-access memory, and an NVIDIA GeForce RTX 2080 Ti graphics card with 11 GB of video memory. The software environment consisted of a Windows 10 Professional operating system. Model training was conducted within the PyCharm2022 integrated development environment, utilizing the PyTorch deep learning framework. CUDA11.2 was employed to accelerate training, thereby optimizing model training speed. During model training, the number of training epochs was set to 400, and the initial learning rate (lr0) was configured to 0.01.

For data acquisition, simultaneously with video data collection, a traditional branch gathering method was employed by five researchers to manually count fruits on each tree. The average of these counts was processed to obtain the orchard ground truth value for the fruits. Subsequently, fruits within the collected videos were counted, and their average was processed to derive the video ground truth value. Concurrently, the number of ID changes, false positives, and false negatives was recorded. The counting result obtained from the estimation system is referred to as the algorithm prediction value (MAX ID).

3. Results and discussions

3.1. Evaluation Metrics

3.1.1. Fruit Detection Experiments

The performance of the trained model was evaluated using metrics including P (precision), recall R (recall), mAP (mean average precision), and computational complexity FLOPs (Floating Point Operations) [34]. The mAP is a comprehensive evaluation metric that reflects the model's detection capability. Average precision values, mAP₅₀ and mAP₅₀₋₉₅, are represented for IoU thresholds of 0.5 and 0.5-0.95, respectively.

3.1.2. Fruit Tracking Experiments

MOTA (Multiple Object Tracking Accuracy) was utilized to represent the algorithm's performance in maintaining tracking, as described by Equation (8). MOTP (Multiple Object Tracking Precision) assessed the matching accuracy between detected and ground truth targets, as defined in Equation (9). Tracking speed FPS (Frame Per Second) was employed to evaluate the algorithm's real-time capability, as indicated in Equation (10). IDSR (ID Switch Rate) refers to the proportion of fruits whose IDs change in the video relative to all counted fruits.

$$\text{MOTA} = 1 - \frac{M}{S} \quad (8)$$

$$\text{MOTP} = \frac{\sum_i d_i}{N} \quad (9)$$

$$\text{FPS} = \frac{1}{T} \quad (10)$$

3.1.3. Fruit Counting Experiments

The differences between predicted and actual observed values were measured using RMSE (Root Mean Square Error) and MAE (Mean Absolute Error), as shown in Equations (11) and (12), respectively. Furthermore, the ACP value (Average Counting Precision) was adopted as the precision metric for citrus fruit counting, as indicated in Equation (13).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (S_i - Y_i)^2} \quad (11)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |S_i - Y_i| \quad (12)$$

$$ACP = 1 - \frac{1}{n} \sum_{i=1}^n \left| \frac{S_i - Y_i}{S_i} \right| \quad (13)$$

3.2. Fruit Image Detection Results

3.2.1. Ablation Study of Object Detection Models

To validate the effectiveness of the improved model, YOLOv8n-SC was compared with YOLOv8n, alongside the commonly utilized Faster RCNN model and the recently introduced lightweight model YOLOv10n. All models were trained and tested using the total dataset, with results presented in Table 1.

Table 1 Comparison of citrus object detection performance among different models

Model	P/%	R/%	mAP ₅₀ /%	mAP ₅₀₋₉₅ /%	Model size /M	FLOPs/G
YOLOv8n	93.7	95.6	90.2	59.1	14.5	8.1
Faster RCNN	88.1	79.3	84.7	56.5	100.3	65.2
YOLOv8n-SC	97.2	97.6	93.7	60.6	14.8	7.9
YOLOv10n	94.5	96.9	90.8	59.9	8.7	6.8

As shown in Table 1, YOLOv8n-SC demonstrated a 1.5% improvement in the average precision metric mAP₅₀₋₉₅ compared to YOLOv8n. Despite a slight increase in YOLOv8n-SC model size due to the new output detection head, and no reduction in model parameters, its computational load marginally decreased. This indicates the efficacy of the model improvements. A significant advantage was observed for YOLOv8n-SC over the traditional two-stage Faster R-CNN model, with a 4.1% improvement in the mAP₅₀₋₉₅ metric. When compared with YOLOv10n, although it had disadvantages in model size and inference speed, YOLOv8n-SC achieved 0.7% higher mAP₅₀₋₉₅ and recall rates. This demonstrates YOLOv8n-SC's strong competitiveness in identifying citrus at the enlargement stage.

3.2.2. Ablation Study of the Object Detection Model

To validate the improved performance of the YOLOv8n model, ablation experiments were conducted on each modified model and the original model using the total dataset and under the same hardware environment, as shown in Table 2.

Table 2 Results of ablation experiment

Model	Detect	CBAM	WIoU	R/%	mAP ₅₀ /%	mAP ₅₀₋₉₅ /%	FLOPs/G
YOLOv8n				95.6	90.2	59.1	8.1
YOLOv8n	√			96.2	92.6	60.2	8.4
YOLOv8n	√	√		96.5	93.3	59.8	8.4

YOLOv8n	√	√	√	97.6	93.7	60.6	7.9
---------	---	---	---	------	------	------	-----

From Table 2, it is observed that after a new detection head was added to the YOLOv8n model, the R and mAP₅₀₋₉₅ for citrus fruits increased by 0.6% and 0.9% respectively. This indicates that the new detection head can improve the recognition rate of minuscule citrus targets. When the C2f module was replaced with C2f_CBAM, the model's detection performance was enhanced, with R and mAP₅₀₋₉₅ improving by 0.3% and 0.7% respectively. This improvement is attributed to CBAM being a lightweight, universal module, which minimally impacts computational load. A slight improvement in model performance was also observed when the loss function was changed to WIoU. A comprehensive comparison of the four convolutional network evaluation indicators demonstrates that YOLOv8n-SC (row 4) exhibited the best performance.

3.2.3. Performance Analysis Under Varied Lighting Conditions

Significant variations in citrus fruit color and texture features are observed under differing light intensities, rendering fruit detail identification challenging. To ascertain the impact of illumination conditions on citrus object detection accuracy, the YOLOv8n-SC model was trained and tested using citrus data acquired under Sunny conditions, Overcast conditions, and Artificial illumination. The results are presented in Table 3.

Table 3 Comparison of citrus detection results under different light conditions

light conditions	P/%	R/%	mAP ₅₀ /%	mAP ₅₀₋₉₅ /%
Sunny conditions	86.1	81.3	83.5	45.1
Overcast conditions	93.5	93.4	96.1	61.4
Artificial illumination	97.2	96.5	98.2	66.9

From Table 3, it is evident that all evaluation metrics were lower under Sunny conditions. This was attributed to reflections, shadows, and fluctuations in light intensity caused by solar radiation, which severely compromised the detection algorithm's performance. Under Overcast conditions, natural light was relatively uniform, enabling more effective target identification by the detection algorithm. Specifically, the P value was increased by 7.4%, the R value by 12.1%, and mAP₅₀₋₉₅ by 16.3%. Under Artificial illumination, all evaluation metrics achieved their optimal state, with mAP₅₀ notably reaching 98.2%. Compared to Overcast conditions, precision was enhanced by 3.7%, recall by 3.1%, and mAP₅₀₋₉₅ by 5.4%. Experimental results indicate that uniform Artificial illumination emphasized the contrast between citrus and the background, simultaneously eliminating image noise attributed to uneven natural light.

3.3. Video-based Fruit Yield Estimation Results

3.3.1. Multi-view Video Tracking Analysis

To minimize estimation errors caused by dense citrus tree branches and fruit occlusion clustering, three sets of videos were captured under Artificial illumination, specifically using Overhead shooting, Horizontal shooting, and Low-angle shooting. Tracking and counting training was performed based on the YOLOv8n-SC model combined with the DeepSORT tracking algorithm, with the confidence score set at 0.7. Tracking results are presented in Table 4.

Table 4 Comparison of citrus tracking results from different shooting angles

Camera angle	MOTA/%	MOTP/%	FPS/s	IDSr/%
Overhead shooting	84.9	95.7	21.74	5.6%
Horizontal shooting	90.8	94.0	21.74	11.8%
Low-angle shooting	85.8	87.9	21.74	27.4%

As shown in Table 4, Horizontal shooting yielded the best detection performance. This is attributed to a reduction in video blind spots and comprehensive capture of fruits across various crown elevation angles. MOTA reached 90.8% and MOTP attained 94.0%. The primary source of error is likely the high ID change rate, observed at 11.8%, attributed to mutual occlusion. Overhead shooting demonstrated the second-best detection performance, characterised by high multi-object tracking precision and stable ID changes. The reason for its lower detection precision compared to Horizontal shooting is likely severe occlusion of lower tree fruits, leading to omissions. Low-angle shooting exhibited the lowest detection performance, as a significant proportion of fruits are located within the canopy region, and occlusion during Low-angle detection diminished the detection rate.

Figure 8 presents result examples. In the Low-angle shooting video shown in Figure 8a, fruits numbered 16, 17, and 30 were continuously tracked at frame 34 but were lost at frame 197. As the disappeared targets were not re-matched within 90 frames (max_age), new IDs (92, 93, 94) were erroneously assigned when these targets reappeared at frame 288. As depicted in Figure 8b, fruit number 47 was occluded by the tree trunk during its movement and was not matched when the confidence score fell below 0.7. However, the original ID was still successfully re-matched after a target loss lasting 87 frames. Figure 8c illustrates that, despite changes in the relative positions of fruits 4, 6, and 11 during the rotation of the viewpoint, no errors occurred in the fruit matching process. This demonstrates the robust tracking capability of the algorithm employed.

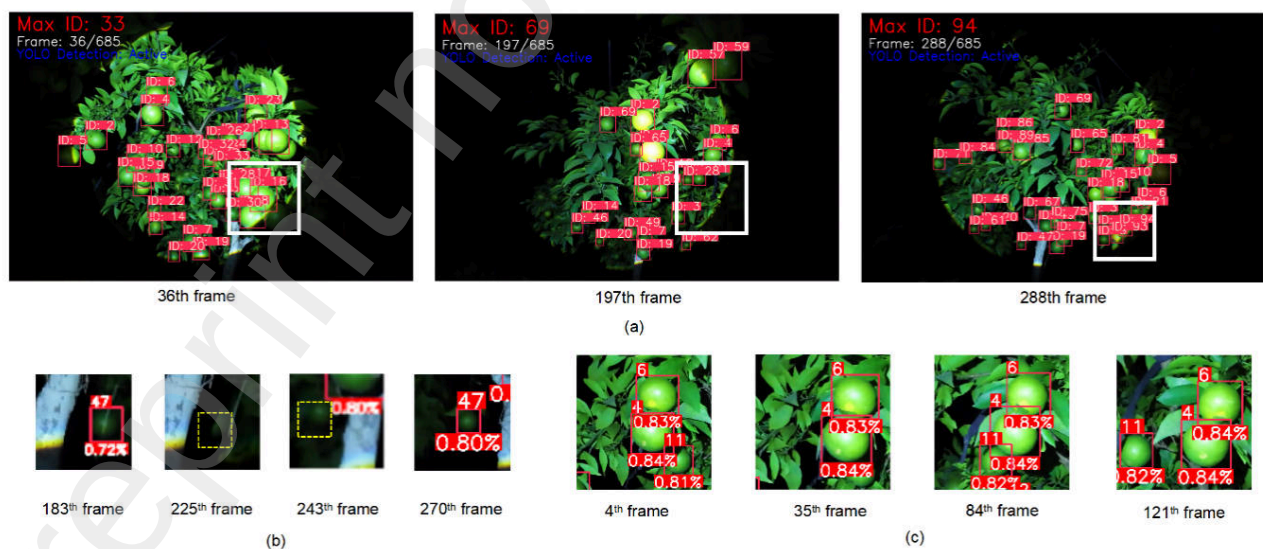


Fig.8-Examples of fruit tracking results under different detection conditions.(a) Example of ID change in elevation Angle.(b) Example of rematching after detecting loss.(c) Example of tracking for intermittent detection of citrus

3.3.2. Multi-angle Video Counting Analysis

Counting accuracy results were obtained based on orchard true values, video true values, and algorithm predicted values, as presented in Table 5.

Table 5 Comparison of citrus counting accuracy from different shooting angles. The required calculations are based on the Video Ground Truth, with the Orchard Ground Truth used for reference only.

Camera angle	Orchard Ground Truth	Video Ground Truth	Algorithm Prediction Value	RMSE	MAE	ACP/%
Overhead shooting		697	539	11.35	9.97	88.87
Horizontal shooting	2195	2103	2298	10.83	9.98	89.46
Low-angle shooting		2091	2484	22.16	20.01	81.03

From the data presented in Table 5, it was observed that citrus fruits during the enlargement stage often exhibit a downward orientation due to increased weight. During Overhead shooting, significant discrepancies were observed between video true values and orchard true values, primarily attributable to occlusion by leaves. This renders such data impractical for reliable citrus yield estimation. During Horizontal shooting, RMSE and MAE values were respectively 11.33 and 10.03 lower than those obtained from Low-angle shooting, whilst ACP was 8.43% higher. This suggests that under Artificial illumination, yield estimation via horizontal video stream capture provides results more closely aligned with the actual number of fruit trees in the orchard.

3.3.3. Tracking Algorithm Comparison

To further validate the algorithm's effectiveness, various citrus counting methods were established by integrating the YOLOv8n-SC model with DeepSORT, SORT, and Tracktor algorithm trackers, respectively. Detection results are presented in Table 6.

Table 6 Comparison of citrus counting accuracy of different tracking algorithms

Model	RMSE	MAE	FPS/s	ACP/%
YOLOv8n-SC+SORT	17.50	16.33	31.84	84.81
YOLOv8n-SC+Tracktor	11.21	10.67	25.25	86.37
YOLOv8n-SC+DeepSORT	10.83	9.98	21.74	89.46

As evidenced by Table 6, the DeepSORT tracking algorithm demonstrated the highest Average Counting Precision. It surpassed SORT and Tracktor trackers by 4.65% and 3.09% respectively. Its Root Mean Square Error and Mean Absolute Error were recorded as 10.83 and 9.98 respectively. However, higher computational complexity and the lowest Frames Per Second were observed, attributed to the necessity of extracting and computing appearance features for every frame.

4. Conclusions

The following outlines the main conclusions derived from the experiment, which proposed the YOLOv8n-SC model integrated with the DeepSORT algorithm for citrus fruit detection and counting during their enlargement stage, exploring estimation accuracy under varying illumination conditions and shooting angles.

(1) The improved YOLOv8n-SC model demonstrated superior comprehensive detection performance compared to YOLOv8n, Faster RCNN, and YOLOv10n models. This substantiates the effectiveness of the model enhancements and its capability for citrus fruit yield estimation during the enlargement stage.

(2) Illumination conditions significantly influence citrus yield estimation. Strong light on Sunny conditions resulted in reflections and shadows, which reduced detection accuracy. Under Overcast conditions, with more uniform illumination, all detection metrics exhibited notable improvements. Artificial illumination provided optimal detection performance, achieving an mAP_{50} of 98.2%, by effectively highlighting fruits and minimizing background interference.

(3) Video shooting angles exerted a significant impact on citrus yield estimation precision. Horizontal shooting video streams demonstrated optimal tracking performance, achieving a MOTA of 90.8% and a MOTP of 94.0%. Overhead shooting video streams were found to contain an insufficient number of visible targets due to fruit occlusion from the limited perspective, thereby compromising the reliability of counting results. Conversely, Low-angle shooting video streams were deemed unsuitable for citrus estimation due to severe ID switching issues and lower overall estimation capability.

(4) Among the evaluated multi-object tracking algorithms for citrus fruits, the YOLOv8n-SC + DeepSORT model combination exhibited the highest estimation precision, with a counting accuracy of 89.46%. This represents an improvement of 4.65% and 3.09% in counting accuracy when compared to YOLOv8n-SC integrated with SORT or Tracktor, respectively.

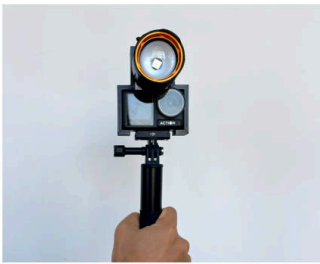
References

- [1] Zhao, P., Wu, X., Cheng, H., Gao, X., Liu, Z., Lai, Q., & Li, H. Performance evaluation of a high-speed maize seed-metering device using an improved YOLOv5s object detection and tracking algorithm. *Smart Agric. Technol.* 11 (2025) DOI: (<https://doi.org/10.1016/j.atech.2025.100997>)
- [2] Zaitoun, N. M., & Aqel, M. J. Survey on Image Segmentation Techniques. *Procedia Comput. Sci.* 797–806 (2015) DOI: 10.32604/cmc.2023.032757
- [3] Akbulut, Y., Guo, Y., Şengür, A., & Aslan, M. An effective color texture image segmentation algorithm based on hermite transform. *Appl. Soft Comput.* 67, 494–504 (2018) DOI: (<https://doi.org/10.1016/j.asoc.2018.03.018>)
- [4] Moussaid, A., El Fkihi, S., Zennayi, Y., Kassou, I., Bourzeix, F., Lahlou, O., ... Imani, Y. Citrus yield prediction using deep learning techniques: A combination of field and satellite data. *J. Open Innov. Technol. Mark. Complex.* 100075 (2023) DOI: (<https://doi.org/10.1016/j.joitmc.2023.100075>)
- [5] Huang, Y., Qian, Y., Wei, H., Lu, Y., Ling, B., & Qin, Y. A survey of deep learning-based object detection methods in crop counting. *Comput. Electron. Agric.* 215, 108425 (2023) DOI: (<https://doi.org/10.1016/j.compag.2023.108425>)

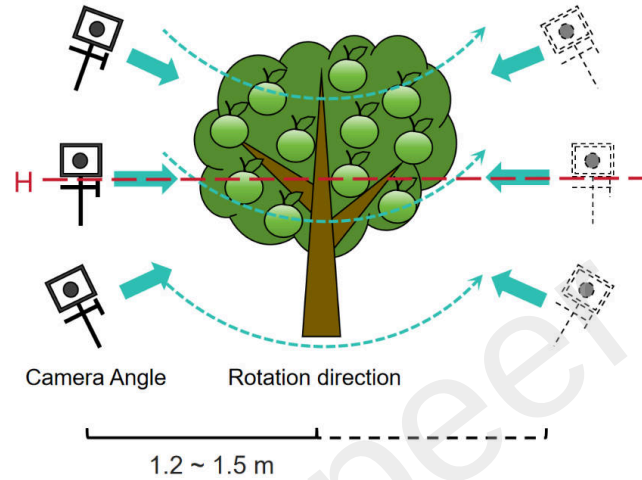
- [6] Xiuling, Z., Huijuan, W., Yu, S., Gang, C., Suhua, Z., & Quanbo, Y. Starting from the structure: A review of small object detection based on deep learning. *Image Vis. Comput.* 146, 105054 (2024) DOI: (<https://doi.org/10.1016/j.imavis.2024.105054>)
- [7] Lu, J., Lee, W. S., Gan, H., & Hu, X. Immature citrus fruit detection based on local binary pattern feature and hierarchical contour analysis. *Biosyst. Eng.* 78-90 (2018) DOI: (<https://doi.org/10.1016/j.biosystemseng.2018.04.009>)
- [8] Dimas Firmanda Al Riza, Lucky Candra Musahada, Romzi Izzudin Aufa, Mochamad Bagus Hermanto, Hermawan Nugroho, & Yusuf Hendrawan. Comparative study of citrus fruits (*Citrus reticulata* Blanco cv. Batu 55) detection and counting with single and double labels based on convolutional neural network using YOLOv7. *Smart Agric. Technol.* 10, 100763 (2025) DOI: (<https://doi.org/10.1016/j.atech.2024.100763>)
- [9] Lu, J., Chen, P., Yu, C., Lan, Y., Yu, L., Yang, R., Niu, H., Chang, H., Yuan, J., & Wang, L. Lightweight green citrus fruit detection method for practical environmental applications. *Comput. Electron. Agric.* 215, 108205 (2023) DOI: (<https://doi.org/10.1016/j.compag.2023.108205>)
- [10] Wu, Z., Sun, X., Jiang, H., & et al. NDMFCS: An automatic fruit counting system in modern apple orchard using abatement of abnormal fruit detection. *Comput. Electron. Agric.* 211, 108036 (2023) DOI: (<https://doi.org/10.1016/j.compag.2023.108036>)
- [11] Lyu, S., Li, R., Zhao, Y., Li, Z., Fan, R., & Liu, S. Green Citrus Detection and Counting in Orchards Based on YOLOv5-CS and AI Edge System. *Sensors.* 576 (2022) DOI: (<https://doi.org/10.3390/s22020576>)
- [12] Liong, S.-T., See, J., Wong, K., & Phan, R. C.-W. Less is more: Micro-expression recognition from video using apex frame. *Signal Process. Image Commun.* 82–92 (2018) DOI: (<https://doi.org/10.1016/j.image.2017.11.006>)
- [13] Yuan, Y., & Liu, Y. BEVEFNet: A multiple object tracking model based on LiDAR-camera fusion. *Procedia Comput. Sci.* 242, 560-567 (2024) DOI: (<https://doi.org/10.1016/j.procs.2024.08.106>)
- [14] Chen, Y., Bin, J., & Kang, C. Application of machine vision and convolutional neural networks in discriminating tobacco leaf maturity on mobile devices. *Smart Agric. Technol.* 5, 100322 (2023) DOI: (<https://doi.org/10.1016/j.atech.2023.100322>)
- [15] Wang, Z., Tao, K., Yuan, J., & Liu, X. Design and experiment on mechanized batch harvesting of Shiitake mushrooms. *Comput. Electron. Agric.* 217, 108593 (2024) DOI: (<https://doi.org/10.1016/j.compag.2023.108593>)
- [16] Wang, Walsh, & Koirala. Mango Fruit Load Estimation Using a Video Based MangoYOLO—Kalman Filter—Hungarian Algorithm Method. *Sensors.* 2742 (2019) DOI: (<https://doi.org/10.3390/s19122742>)
- [17] Jin, S., Zhou, L., & Zhou, H. CO-YOLO: A lightweight and efficient model for *Camellia oleifera* fruit object detection and posture determination. *Comput. Electron. Agric.* 235, 110394 (2025) DOI: (<https://doi.org/10.1016/j.compag.2025.110394>)
- [18] Li, X., Wang, X., Ong, P., Yi, Z., Ding, L., & Han, C. Fast Recognition and Counting Method of Dragon Fruit Flowers and Fruits Based on Video Stream. *Sensors.* 23(20), 8444 (2023) DOI: (<https://doi.org/10.3390/s23208444>)

- [19] Font, D., Tresanchez, M., Martínez, D., Moreno, J., Clotet, E., & Palacín, J. Vineyard Yield Estimation Based on the Analysis of High Resolution Images Obtained with Artificial Illumination at Night. *Sensors*. 8284–8301 (2015) DOI: (<https://doi.org/10.3390/s150408284>)
- [20] Kurtulmus, F., Lee, W. S., & Vardar, A. Green citrus detection using 'eigenfruit', color and circular Gabor texture features under natural outdoor conditions. *Comput. Electron. Agric.* 140-149 (2011) DOI: (<https://doi.org/10.1016/j.compag.2011.07.001>)
- [21] Sun, H., Ren, R., Zhang, S., Tan, C., & Jing, J. Maturity detection of 'Huping' jujube fruits in natural environment using YOLO-FHLD. *Smart Agric. Technol.* 9, 100670 (2024) DOI: (<https://doi.org/10.1016/j.atech.2024.100670>)
- [22] Tong, K., & Wu, Y. Deep learning-based detection from the perspective of small or tiny objects: A survey. *Image Vis. Comput.* 123, 104471 (2022) DOI: (<https://doi.org/10.1016/j.imavis.2022.104471>)
- [23] Koyun, O. C., Keser, R. K., Akkaya, İ. B., & Töreyn, B. U. Focus-and-Detect: A small object detection framework for aerial images. *Signal Process. Image Commun.* 104, 116675 (2022) DOI: (<https://doi.org/10.1016/j.image.2022.116675>)
- [24] Li, H., Huang, J., Gu, Z., He, D., Huang, J., & Wang, C. Positioning of mango picking point using an improved YOLOv8 architecture with object detection and instance segmentation. *Biosyst. Eng.* 247, 202-220 (2024) DOI: (<https://doi.org/10.1016/j.biosystemseng.2024.09.015>)
- [25] Woo, S., Park, J., Lee, J.-Y., & Kweon, I. S. CBAM: Convolutional Block Attention Module. In *Comput. Vis. - ECCV 2018, Lecture Notes in Computer Science* (pp. 3-19) (2018) DOI: (<https://doi.org/10.48550/arXiv.1807.06521>)
- [26] Jin, Y., Tian, X., Zhang, Z., Liu, P., & Tang, X. C2F: An effective coarse-to-fine network for video summarization. *Image Vis. Comput.* 144, 104962 (2024) DOI: (<https://doi.org/10.1016/j.imavis.2024.104962>)
- [27] Xu, W., Cui, C., Ji, Y., Li, X., & Li, S. YOLOv8-MPEB small target detection algorithm based on UAV images. *Heliyon*. 10(8), e29501 (2024) DOI: (<https://doi.org/10.1016/j.heliyon.2024.e29501>)
- [28] Zheng, Z., Wang, P., Ren, D., Liu, W., Ye, R., Hu, Q., & Zuo, W. Enhancing Geometric Factors in Model Learning and Inference for Object Detection and Instance Segmentation. *IEEE Trans. Cybern.* 8574–8586 (2022) DOI: 10.1109/TCYB.2021.3095305
- [29] Zhang, H., Xu, C., & Zhang, S. Inner-IoU: More Effective Intersection over Union Loss with Auxiliary Bounding Box (2023) DOI: (<https://doi.org/10.48550/arXiv.2311.02877>)
- [30] Tong, Z., Chen, Y., Xu, Z., & Yu, R. Wise-IoU: Bounding Box Regression Loss with Dynamic Focusing Mechanism (2023) DOI: (<https://doi.org/10.48550/arXiv.2301.10051>)
- [31] Xiao, D., Wang, H., Liu, Y., Li, W., & Li, H. DHSW-YOLO: A duck flock daily behavior recognition model adaptable to bright and dark conditions. *Comput. Electron. Agric.* 225, 109281 (2024) DOI: (<https://doi.org/10.1016/j.compag.2024.109281>)
- [32] Tu, S., Huang, Y., Huang, Q., et al. Estimation of passion fruit yield based on YOLOv8n + OC-SORT + CRCM algorithm. *Comput. Electron. Agric.* 229, 109727 (2025) DOI: (<https://doi.org/10.1016/j.compag.2024.109727>)
- [33] He, L., Wu, F., Du, X., et al. Cascade-SORT: A robust fruit counting approach using multiple features cascade matching. *Comput. Electron. Agric.* 107223 (2022) DOI: (<https://doi.org/10.1016/j.compag.2022.107223>)

- 1 [34] Wu, Z., Sun, X., Jiang, H., et al. Twice matched fruit counting system: An automatic fruit counting pipeline in
2 modern apple orchard using mutual and secondary matches. Biosyst. Eng. 234, 140-155 (2023) DOI:
3 (<https://doi.org/10.1016/j.biosystemseng.2023.09.005>)



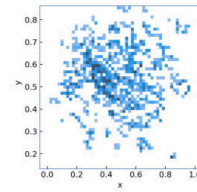
(a)



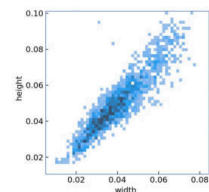
(b)



(c)



(d)



(e)

Sunny conditions

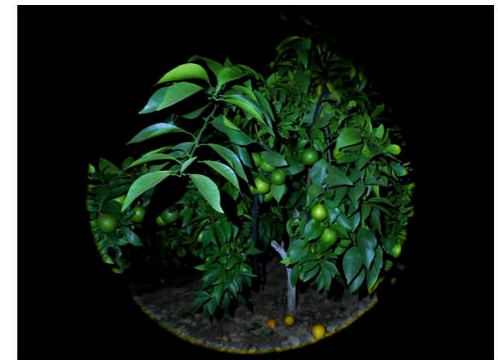
Overcast conditions

Artificial illumination

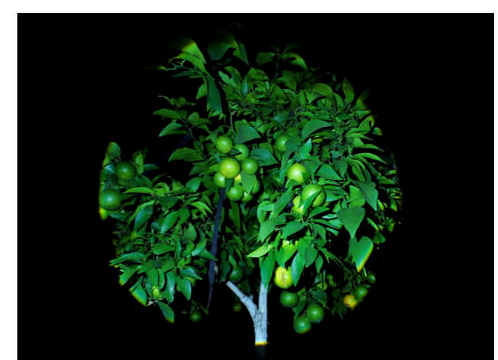
Overhead
shooting

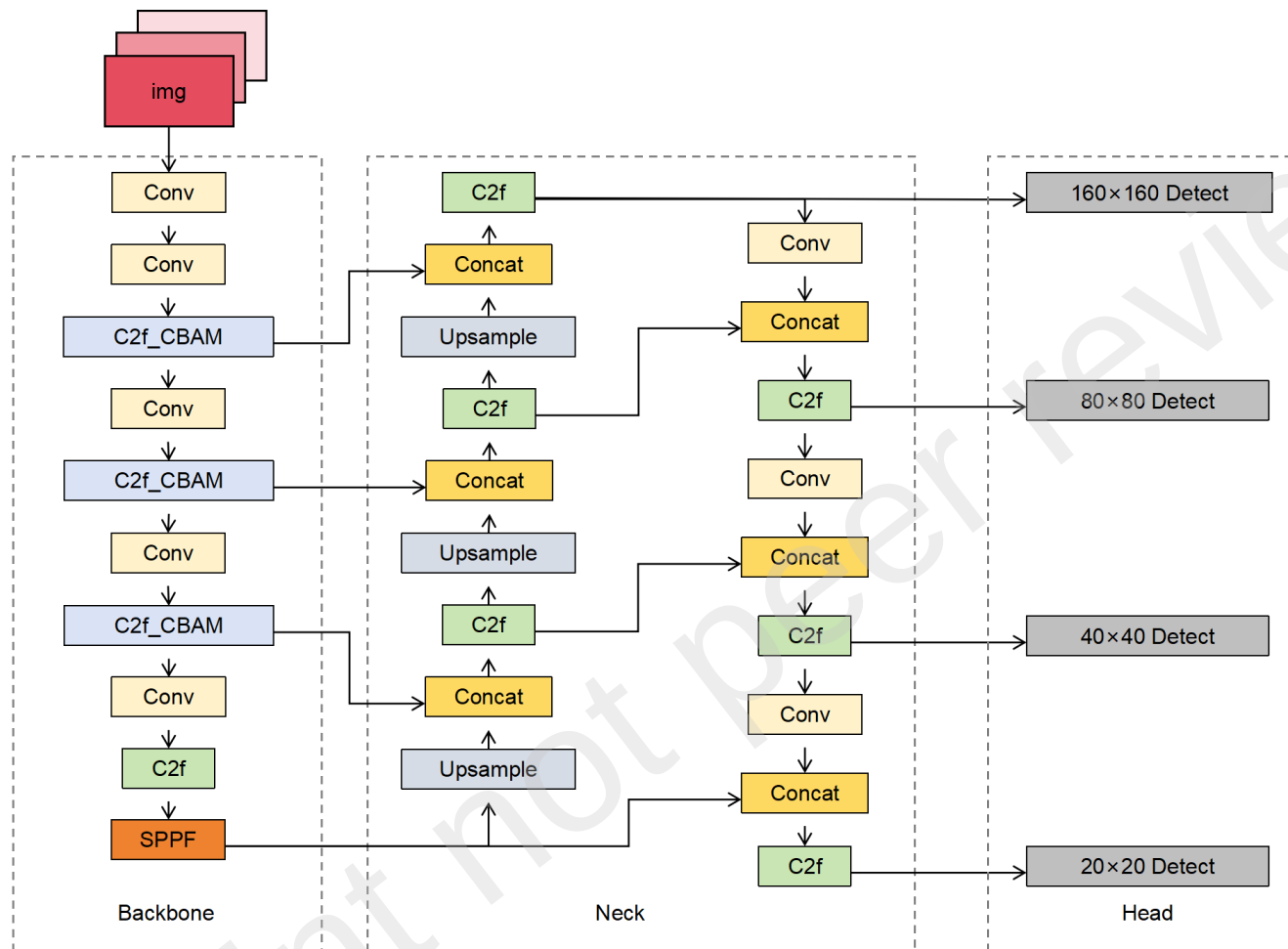


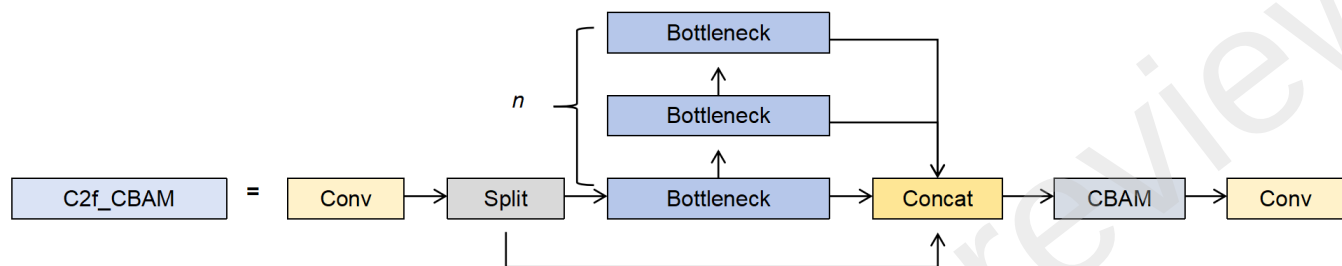
Horizontal
shooting

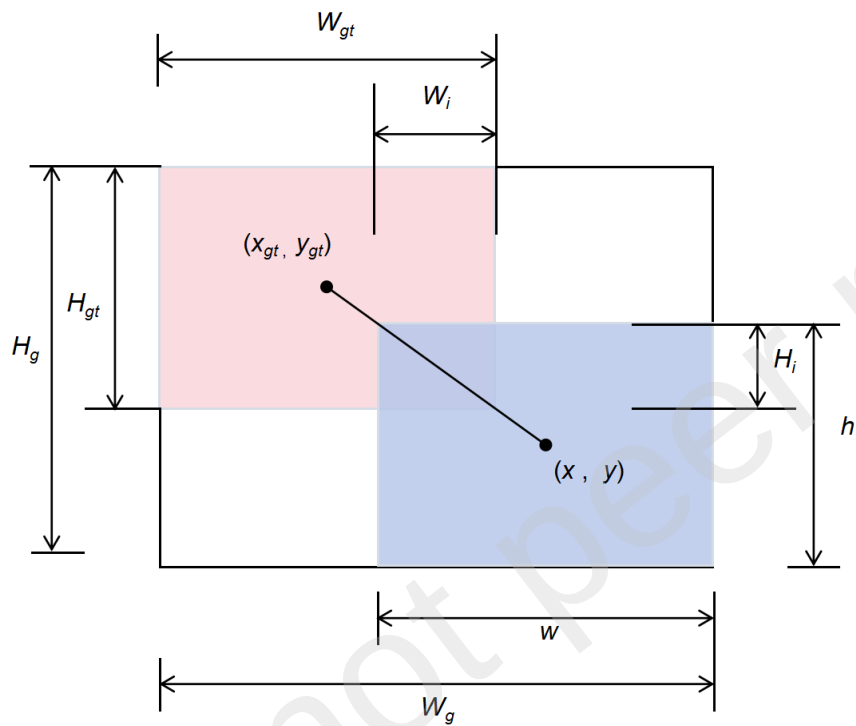


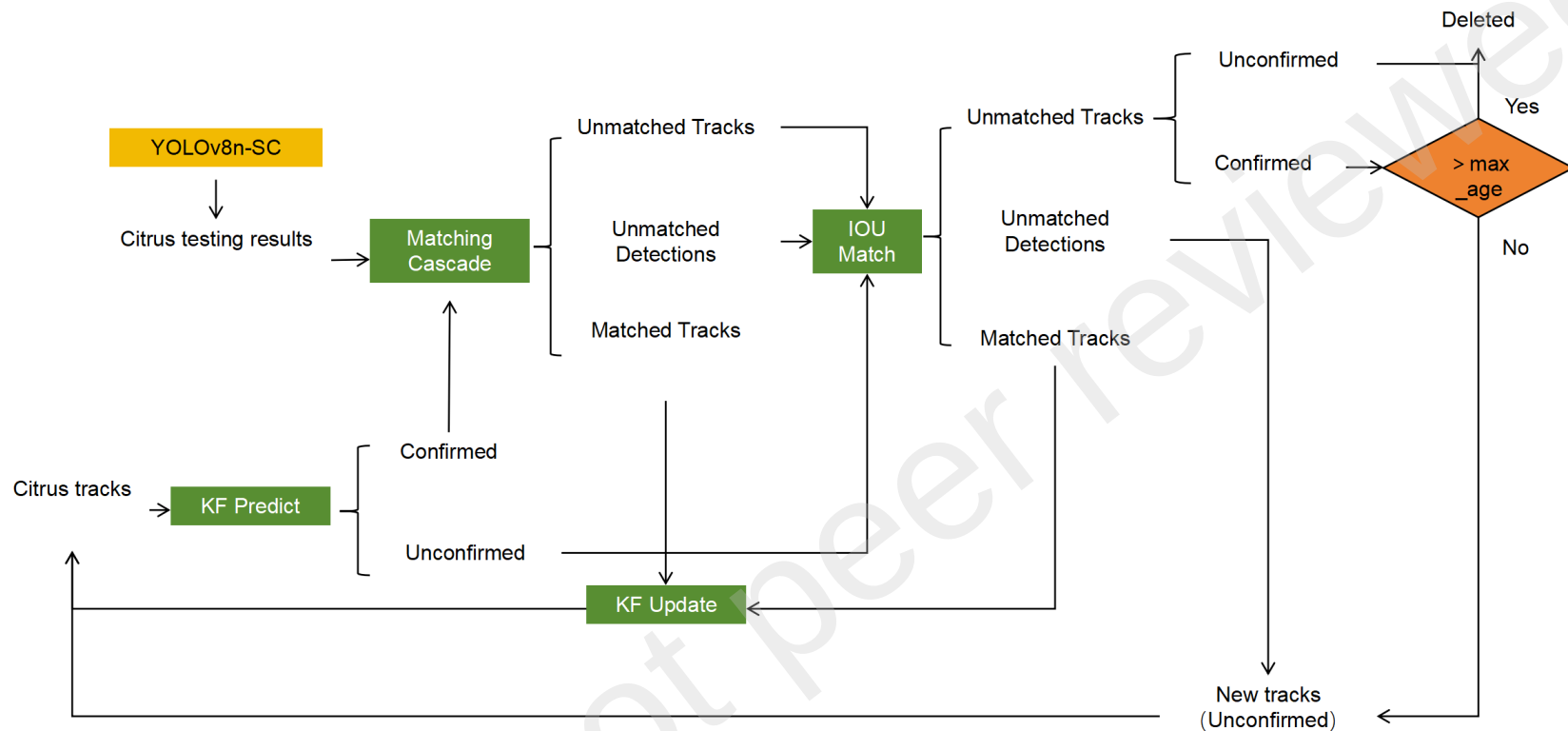
Low-angle
shooting

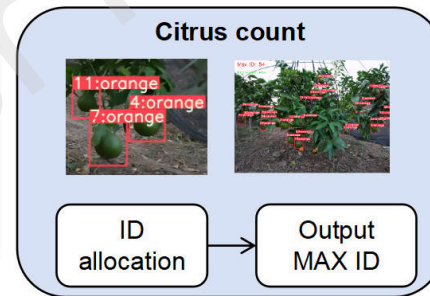
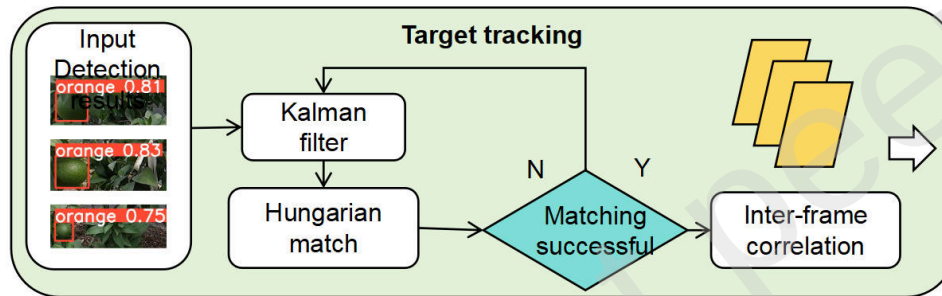
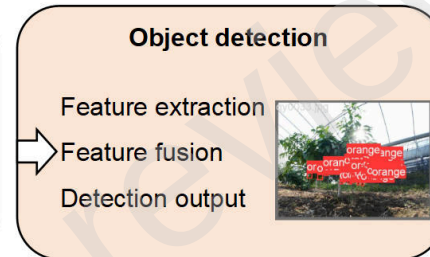
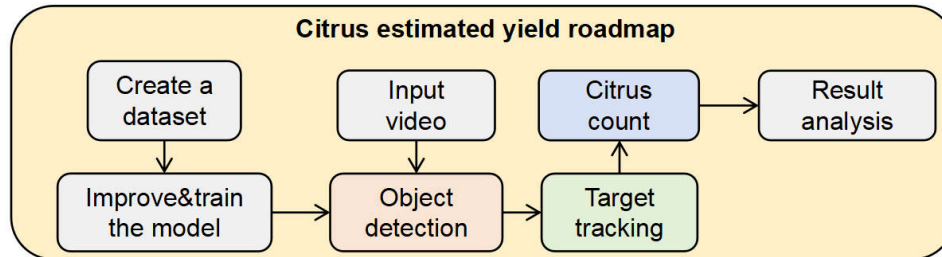






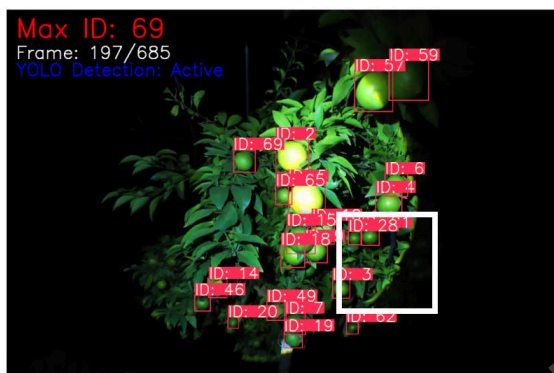








36th frame



197th frame

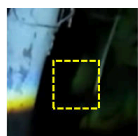


288th frame

(a)



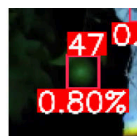
183th frame



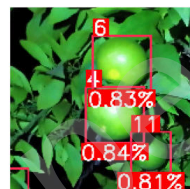
225th frame



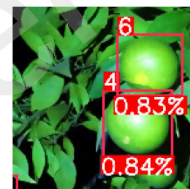
243th frame



270th frame



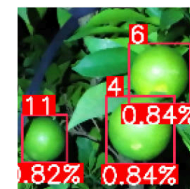
4th frame



35th frame



84th frame



121th frame

(b)

(c)